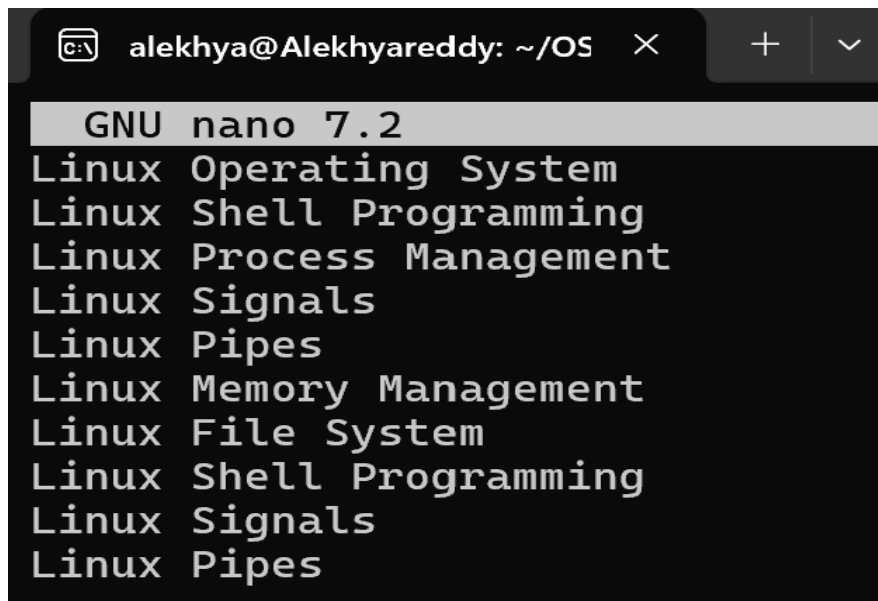


SKILL WEEK 12

1 .

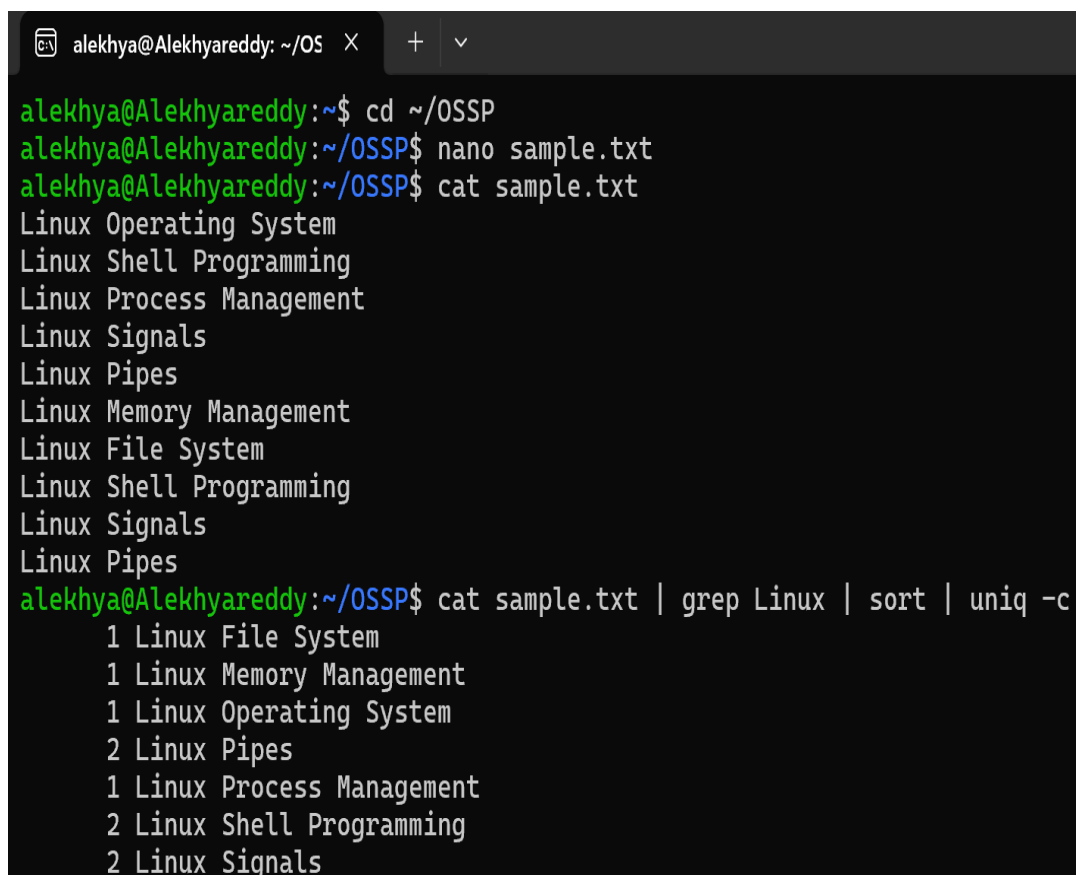
Aim: To execute Linux pipelines, launch multiple background jobs, monitor running processes and system resources, detect job failures, and analyze performance using Linux commands.

Program :



```
alekhya@Alekhyareddy: ~/OS
GNU nano 7.2
Linux Operating System
Linux Shell Programming
Linux Process Management
Linux Signals
Linux Pipes
Linux Memory Management
Linux File System
Linux Shell Programming
Linux Signals
Linux Pipes
```

Output:



```
alekhya@Alekhyareddy:~$ cd ~/OSSP
alekhya@Alekhyareddy:~/OSSP$ nano sample.txt
alekhya@Alekhyareddy:~/OSSP$ cat sample.txt
Linux Operating System
Linux Shell Programming
Linux Process Management
Linux Signals
Linux Pipes
Linux Memory Management
Linux File System
Linux Shell Programming
Linux Signals
Linux Pipes
alekhya@Alekhyareddy:~/OSSP$ cat sample.txt | grep Linux | sort | uniq -c
  1 Linux File System
  1 Linux Memory Management
  1 Linux Operating System
  2 Linux Pipes
  1 Linux Process Management
  2 Linux Shell Programming
  2 Linux Signals
```

```
alekhya@Alekhyareddy:~/OSSP$ sleep 200 &
```

```
[1] 469
```

```
alekhya@Alekhyareddy:~/OSSP$ sleep 250 &
```

```
[2] 473
```

```
alekhya@Alekhyareddy:~/OSSP$ sleep 300 &
```

```
[3] 474
```

```
alekhya@Alekhyareddy:~/OSSP$ jobs -l
```

```
[1] 469 Running sleep 200 &
```

```
[2]- 473 Running sleep 250 &
```

```
[3]+ 474 Running sleep 300 &
```

```
alekhya@Alekhyareddy:~/OSSP$ ps -f
```

```

UID          PID    PPID  C  STIME TTY          TIME CMD
alekhya      302     301  0   05:27 pts/0        00:00:00 -bash
alekhya      469     302  0   05:33 pts/0        00:00:00 sleep 200
alekhya      473     302  0   05:33 pts/0        00:00:00 sleep 250
alekhya      474     302  0   05:33 pts/0        00:00:00 sleep 300
alekhya      475     302  0   05:33 pts/0        00:00:00 ps -f

```

```
alekhya@Alekhyareddy:~/OSSP$ top
```

```
top - 05:34:07 up 7 min, 1 user, load average: 0.06, 0.02, 0.00
```

```
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie
```

```
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni,100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
```

```
MiB Mem : 15743.6 total, 14031.9 free, 571.1 used, 1337.1 buff/cache
```

```
MiB Swap: 4096.0 total, 4096.0 free, 0.0 used. 15172.5 avail Mem
```

```
PID to signal/kill [default pid = 1]
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	21740	13156	9800	S	0.0	0.1	0:00.43	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.0	0:00.00	init-systemd(Ub
6	root	20	0	3196	2128	2008	S	0.0	0.0	0:00.00	init
52	root	19	-1	50368	16240	15112	S	0.0	0.1	0:00.13	systemd-journal
102	root	20	0	25756	6920	5168	S	0.0	0.0	0:00.13	systemd-udev
130	systemd+	20	0	21344	13208	11088	S	0.0	0.1	0:00.06	systemd-resolve
131	systemd+	20	0	91036	7972	7008	S	0.0	0.0	0:00.02	systemd-timesyn
167	root	20	0	4244	2684	2432	S	0.0	0.0	0:00.01	cron
168	message+	20	0	9608	5224	4540	S	0.0	0.0	0:00.03	dbus-daemon
182	root	20	0	17976	8884	7848	S	0.0	0.1	0:00.05	systemd-logind
188	root	20	0	1756628	13572	11444	S	0.0	0.1	0:00.14	wsl-pro-service
204	syslog	20	0	222516	6024	4628	S	0.0	0.0	0:00.05	rsyslogd
207	root	20	0	3124	1972	1836	S	0.0	0.0	0:00.00	agetty
220	root	20	0	107036	23108	13656	S	0.0	0.1	0:00.14	unattended-upgr
300	root	20	0	3184	1112	980	S	0.0	0.0	0:00.00	SessionLeader
301	root	20	0	3200	1248	1108	S	0.0	0.0	0:00.02	Relay(302)
302	alekhya	20	0	6208	5480	3708	S	0.0	0.0	0:00.04	bash
303	root	20	0	6700	4608	3828	S	0.0	0.0	0:00.00	login

alekhya@AlekhyaReddy: ~/OS X + v

alekhya@AlekhyaReddy:~/OSSP\$ kill %2

alekhya@AlekhyaReddy:~/OSSP\$ jobs -l

[1] 469 Running sleep 200 &

[2] 473 Terminated sleep 250

[3]- 474 Running sleep 300 &

[4]+ 476 Stopped (signal) top

alekhya@AlekhyaReddy:~/OSSP\$ time ls -R

```
.:
LICENSE          dynamic_memory  memory_leak     task1_wait.c    task5_process   task_valgrind.c
Makefile          dynamic_memory.c memory_leak.c    task1_wait.c.save task5_process.c  task_valgrind_leak
README.md         error.log       mmap_file       task2_quotes    task_error_handling task_valgrind_leak.c
assets           expected_report.txt mmap_file.c      task2_quotes.c  task_error_handling.c test_report.txt
bin              include         mmap_file.txt   task2_single_quotes task_large_pipeline test_script.sh
calculator.c      input.txt       module_test     task2_single_quotes.c task_large_pipeline.c tests
calculator.h      io_redirect     obj             task3_double_quotes task_process_groups thread_counter
calculator.o      io_redirect.c   output.txt      task3_double_quotes.c task_process_groups.c thread_counter.c
calculator_test   library_copy    producer_consumer task3_fork       task_resume_job   thread_counter_mutex
calculator_test.c library_copy.c  producer_consumer.c task3_fork.c     task_resume_job.c thread_counter_mutex.c
client           library_copy.txt sample.txt       task3_input     task_sigint_forward traditional_io
client.c          lowlevel_copy  scripts         task3_input.c   task_sigint_forward.c traditional_io.c
client_to_server lowlevel_copy.c server           task4_escape    task_sigint_forward.c.save valgrind_report.txt
cow_practical    lowlevel_copy.txt server.c         task4_escape.c  task_signal_actions week4_T2
cow_practical.c  ls_grep        server_to_client task4_history    task_signal_actions.c week4_T2.c
deadlock         ls_grep.c      signal_handler  task4_history.c task_signal_behavior week5_double_quotes.c
deadlock.c       main_loop      signal_handler.c task4_states     task_signal_behavior.c week5double_quotes
deadlock_prevention main_loop.c    source.txt     task4_states.c  task_sigsttp      task_sigsttp.c
deadlock_prevention.c main_module.c  src            task5_memory    task_sigsttp.c    task_valgrind
docs             main_module.o  task1_wait     task5_memory.c  task_valgrind
```

./assets:

architecture.png

./bin:

fork_exec

./docs:

architecture.txt design.md design_decisions.txt execution_flow.txt report.pdf syscalls.txt

./include:

builtin.h executor.h parser.h shell.h

./obj:

builtin.o executor.o main.o parser.o shell.o

./scripts:

run.sh

./src:

builtin.c executor.c fork_exec.c main.c parser.c shell.c

./tests:

Aim: To document the architecture of a Linux program, explain its execution path and system calls, create a flow diagram, record design decisions, and review the program documentation.

```
GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <time.h>
#include <sys/wait.h>

int main()
{
    printf("Program Started\n");

    printf("Process ID : %d\n", getpid());
    printf("Parent Process ID : %d\n", getppid());

    printf("Executing Program...\n");

    sleep(2);

    printf("Program Finished Successfully\n");

    return 0;
}
```

```

alekhya@Alekhyareddy: ~/OSS X + v
alekhya@Alekhyareddy:~$ cd ~/OSSP
alekhya@Alekhyareddy:~/OSSP$ nano skill112_task2.c
alekhya@Alekhyareddy:~/OSSP$ gcc skill112_task2.c -o skill112_task2
alekhya@Alekhyareddy:~/OSSP$ ./skill112_task2
Program Started
Process ID : 540
Parent Process ID : 509
Executing Program...
Program Finished Successfully
alekhya@Alekhyareddy:~/OSSP$ strace ./skill112_task2
execve("./skill112_task2", [ "./skill112_task2" ], 0x7ffe3ae60df0 /* 27 vars */) = 0
brk(NULL)                               = 0x6412d3eaf000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fdb38145000
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=32539, ...}) = 0
mmap(NULL, 32539, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7fdb3813d000
close(3)                                = 0
openat(AT_FDCWD, "/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\3\0\0\0\1\0\0\0\220\243\2\0\0\0\0\0"... , 832) = 832
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"... , 784, 64) = 784
fstat(3, {st_mode=S_IFREG|0755, st_size=2125328, ...}) = 0
pread64(3, "\6\0\0\0\4\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0\0"... , 784, 64) = 784
mmap(NULL, 2170256, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7fdb37e00000
mmap(0x7fdb37e28000, 1605632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x280000) = 0x7fdb37e28000
mmap(0x7fdb37fb0000, 323584, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1b0000) = 0x7fdb37fb0000
mmap(0x7fdb37ffff000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1fe000) = 0x7fdb37ffff000
mmap(0x7fdb38005000, 52624, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7fdb38005000
close(3)                                = 0
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fdb3813a000
arch_prctl(ARCH_SET_FS, 0x7fdb3813a740) = 0
set_tid_address(0x7fdb3813aa10)         = 544

```

```

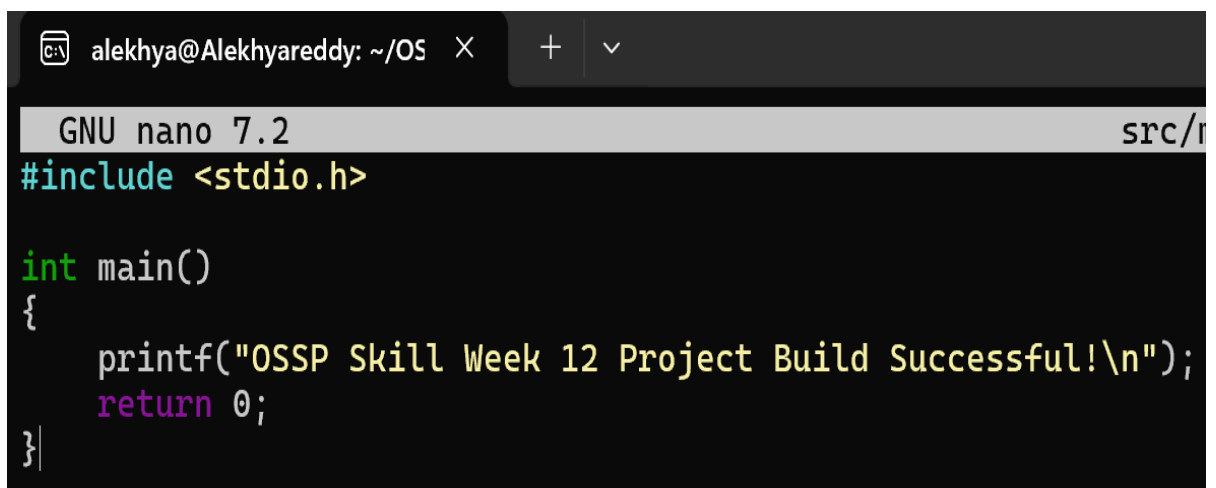
mmap(NULL, 12288, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7fdb3813a000
arch_prctl(ARCH_SET_FS, 0x7fdb3813a740) = 0
set_tid_address(0x7fdb3813aa10) = 544
set_robust_list(0x7fdb3813aa20, 24) = 0
rseq(0x7fdb3813b060, 0x20, 0, 0x53053053) = 0
mprotect(0x7fdb37fff000, 16384, PROT_READ) = 0
mprotect(0x6412af794000, 4096, PROT_READ) = 0
mprotect(0x7fdb38185000, 8192, PROT_READ) = 0
prlimit64(0, RLIMIT_STACK, NULL, {rlim_cur=8192*1024, rlim_max=RLIM64_INFINITY}) = 0
munmap(0x7fdb3813d000, 32539) = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(0x88, 0), ...}) = 0
getrandom("\xa5\x02\x66\xb7\xff\x21\xa2\xcc", 8, GRND_NONBLOCK) = 8
brk(NULL) = 0x6412d3eaf000
brk(0x6412d3ed0000) = 0x6412d3ed0000
write(1, "Program Started\n", 16Program Started
) = 16
getpid() = 544
write(1, "Process ID : 544\n", 17Process ID : 544
) = 17
getppid() = 541
write(1, "Parent Process ID : 541\n", 24Parent Process ID : 541
) = 24
write(1, "Executing Program...\n", 21Executing Program...
) = 21
clock_nanosleep(CLOCK_REALTIME, 0, {tv_sec=2, tv_nsec=0}, 0x7ffeb9088e90) = 0
write(1, "Program Finished Successfully\n", 30Program Finished Successfully
) = 30
exit_group(0) = ?
+++ exited with 0 +++

```

3 .

Aim: To organize source code into separate folders, create a build script, add a README file, verify the repository structure, and validate the project deliverables in Linux.

Program:



```

GNU nano 7.2 src/r
#include <stdio.h>

int main()
{
    printf("OSSP Skill Week 12 Project Build Successful!\n");
    return 0;
}

```

```
alekhya@Alekhyareddy: ~/OS  ×  +
GNU nano 7.2
#ifdef CALCULATOR_H
#define CALCULATOR_H

int add(int a, int b);
int subtract(int a, int b);

#endif
```

```
alekhya@Alekhyareddy: ~/OS  ×  +  v
GNU nano 7.2 docs/README.md
# OSSP Skill Week 12

## Project
Operating System and Systems Programming

## Description
This project demonstrates Linux system programming concepts using C.

## Build
The project is compiled using GCC.

## Run
The generated executable is used to test the project.

## Structure
src      - Source files
include  - Header files
docs     - Documentation
```

```
alekhya@Alekhyareddy: ~/OS × + v
GNU nano 7.2
#!/bin/bash

echo "Building OSSP Project..."

gcc src/main.c -o project_build

if [ $? -eq 0 ]; then
    echo "Build Completed Successfully."
else
    echo "Build Failed."
fi
```

```
alekhya@Alekhyareddy: ~/OS × + v
GNU nano 7.2
OSSP SKILL WEEK 12 REPORT

Project Build and Repository Organization

Source Directory:
src/

Header Directory:
include/

Documentation Directory:
docs/

Build Script:
build.sh

Executable:
project_build

Build Status:
Successful
```

Output:

```
alekhya@Alekhyareddy: ~/OS X + v
alekhya@Alekhyareddy:~$ cd ~/OSSP
alekhya@Alekhyareddy:~/OSSP$ mkdir -p src include docs
alekhya@Alekhyareddy:~/OSSP$ touch src/main.c
alekhya@Alekhyareddy:~/OSSP$ touch include/calculator.h
alekhya@Alekhyareddy:~/OSSP$ touch docs/README.md
alekhya@Alekhyareddy:~/OSSP$ ls -R
.:
LICENSE                include                producer_consumer    task3_input          task_signal_actions.c
Makefile               input.txt             producer_consumer.c  task3_input.c        task_signal_behavior
README.md              io_redirect           sample.txt           task4_escape         task_signal_behavior.c
assets                 io_redirect.c         scripts              task4_escape.c       task_sigsttp
bin                    library_copy          server               task4_history        task_sigsttp.c
calculator.c            library_copy.c        server.c             task4_history.c      task_valgrind
calculator.h            library_copy.txt      server_to_client     task4_states         task_valgrind.c
calculator.o            lowlevel_copy        signal_handler       task4_states.c       task_valgrind_leak
calculator_test         lowlevel_copy.c      skill12_task2       task5_memory         task_valgrind_leak.c
calculator_test.c      lowlevel_copy.txt    skill12_task2.c     task5_memory.c       test_report.txt
client                 ls_grep              source.txt           task5_process        test_script.sh
client.c               ls_grep.c            task1_wait           task5_process.c      tests
client_to_server       main_loop             src                 task_error_handling  thread_counter
cow_practical          main_loop.c           task1_wait.c         task_error_handling.c thread_counter.c
cow_practical.c        main_module.c         task1_wait.c.save   task_large_pipeline  thread_counter_mutex
deadlock               main_module.o         task2_quotes        task_large_pipeline.c thread_counter_mutex.c
deadlock.c             memory_leak           task2_quotes.c      task_process_groups  traditional_io
deadlock_prevention    memory_leak.c         task2_single_quotes task_process_groups.c traditional_io.c
deadlock_prevention.c  mmap_file             task2_single_quotes task_resume_job       valgrind_report.txt
docs                   mmap_file.c           task3_double_quotes task_resume_job.c    week4_T2
dynamic_memory         mmap_file.txt         task3_double_quotes task_sigint_forward  week4_T2.c
dynamic_memory.c       module_test           task3_double_quotes task_sigint_forward.c week_5_double_quotes
error.log              obj                   task3_fork          task_sigint_forward.c.save week_5double_quotes
expected_report.txt    output.txt            task3_fork.c        task_signal_actions
```

```
./tests:
test_executor.c test_parser.c
alekhya@Alekhyareddy:~/OSSP$ nano src/main.c
alekhya@Alekhyareddy:~/OSSP$ nano include/calculator.h
alekhya@Alekhyareddy:~/OSSP$ nano docs/README.md
alekhya@Alekhyareddy:~/OSSP$ nano build.sh
alekhya@Alekhyareddy:~/OSSP$ chmod +x build.sh
alekhya@Alekhyareddy:~/OSSP$ ./build.sh
Building OSSP Project...
Build Completed Successfully.
alekhya@Alekhyareddy:~/OSSP$ ./project_build
OSSP Skill Week 12 Project Build Successful!
alekhya@Alekhyareddy:~/OSSP$ nano report.txt
alekhya@Alekhyareddy:~/OSSP$ tree
```

```
.
├── LICENSE
├── Makefile
├── README.md
├── assets
│   └── architecture.png
├── bin
│   └── fork_exec
├── build.sh
├── calculator.c
├── calculator.h
├── calculator.o
├── calculator_test
├── calculator_test.c
├── client
├── client.c
├── client_to_server
├── cow_practical
├── cow_practical.c
├── deadlock
├── deadlock.c
├── deadlock_prevention
├── deadlock_prevention.c
├── docs
│   ├── README.md
│   ├── architecture.txt
│   ├── design.md
│   ├── design_decisions.txt
│   ├── execution_flow.txt
│   ├── report.pdf
│   └── syscalls.txt
├── dynamic_memory
├── dynamic_memory.c
├── error.log
├── expected_report.txt
├── include
│   └── builtin.h
```



alekhya@Alekhyareddy: ~/OS



```
— task4_escape.c
— task4_history
— task4_history.c
— task4_states
— task4_states.c
— task5_memory
— task5_memory.c
— task5_process
— task5_process.c
— task_error_handling
— task_error_handling.c
— task_large_pipeline
— task_large_pipeline.c
— task_process_groups
— task_process_groups.c
— task_resume_job
— task_resume_job.c
— task_sigint_forward
— task_sigint_forward.c
— task_sigint_forward.c.save
— task_signal_actions
— task_signal_actions.c
— task_signal_behavior
— task_signal_behavior.c
— task_sigtstp
— task_sigtstp.c
— task_valgrind
— task_valgrind.c
— task_valgrind_leak
— task_valgrind_leak.c
— test_report.txt
— test_script.sh
— tests
  |— test_executor.c
  |— test_parser.c
— thread_counter
— thread_counter.c
— thread_counter_mutex
— thread_counter_mutex.c
— traditional_io
— traditional_io.c
— valgrind_report.txt
— week4_T2
— week4_T2.c
— week_5_double_quotes.c
— week_5double_quotes
```

9 directories, 142 files

alekhya@Alekhyareddy:~/OSSP\$ |

4 .

Aim: To create demo scenarios, validate Linux features, demonstrate pipelines and signals, verify program execution, and perform a final project review in Linux.

Program :

```
alekhya@Alekhyareddy: ~/OS
GNU nano 7.2
Linux Memory
Linux Operating System
Linux Pipes
Linux Processes
Linux Shell
Linux Signals
Linux Signals
Linux Pipes|
```

Output:

```
alekhya@Alekhyareddy: ~/OS
alekhya@Alekhyareddy:~$ cd ~/OSSP
alekhya@Alekhyareddy:~/OSSP$ nano demo.txt
alekhya@Alekhyareddy:~/OSSP$ cat demo.txt | grep Linux | sort | uniq -c
    1 Linux Memory
    1 Linux Operating System
    2 Linux Pipes
    1 Linux Processes
    1 Linux Shell
    2 Linux Signals
alekhya@Alekhyareddy:~/OSSP$ sleep 120 &
[1] 682
alekhya@Alekhyareddy:~/OSSP$ jobs -l
[1]+  682 Running                  sleep 120 &
alekhya@Alekhyareddy:~/OSSP$ kill -STOP PID
-bash: kill: PID: arguments must be process or job IDs
alekhya@Alekhyareddy:~/OSSP$ kill -STOP 7212
-bash: kill: (7212) - No such process
alekhya@Alekhyareddy:~/OSSP$ jobs -l
[1]+  682 Running                  sleep 120 &
alekhya@Alekhyareddy:~/OSSP$ jobs -l
[1]+  682 Running                  sleep 120 &
alekhya@Alekhyareddy:~/OSSP$ kill %1
alekhya@Alekhyareddy:~/OSSP$ jobs -l
[1]+  682 Terminated              sleep 120
```

```

alekhya@AlekhyaReddy: ~/OS  ×  +  ▾
[1]+  682 Terminated                sleep 120
alekhya@AlekhyaReddy:~/OSSP$ ls -R
.:
LICENSE                expected_report.txt  producer_consumer    task3_fork.c          task_signal_actions.c
Makefile               include              producer_consumer.c  task3_input           task_signal_behavior
README.md              input.txt            project_build         task3_input.c         task_signal_behavior.c
assets                 io_redirect          report.txt            task4_escape          task_sigstsp
bin                    io_redirect.c        sample.txt            task4_escape.c        task_sigstsp.c
build.sh               library_copy         scripts              task4_history         task_valgrind
calculator.c            library_copy.c       server               task4_history.c        task_valgrind.c
calculator.h            library_copy.txt     server.c             task4_states          task_valgrind_leak
calculator.o            lowlevel_copy       signal_handler.c     task4_states.c        task_valgrind_leak.c
calculator_test         lowlevel_copy.c     skill12_task2       task5_memory          test_report.txt
calculator_test.c       lowlevel_copy.txt   skill12_task2.c     task5_memory.c        test_script.sh
client                 ls_grep             source.txt           task5_process         tests
client.c               ls_grep.c           task1_wait           task5_process.c       thread_counter
client_to_server        main_loop           task1_wait.c         task_error_handling   thread_counter.c
cow_practical          main_module.c       task1_wait.c.save    task_large_pipeline   thread_counter_mutex
cow_practical.c         main_module.o       task2_quotes         task_large_pipeline.c traditional_io
deadlock               memory_leak          task2_quotes.c       task_process_groups   traditional_io.c
deadlock_prevention    mmap_file            task2_single_quotes  task_process_groups.c valgrind_report.txt
deadlock_prevention.c  mmap_file.c         task3_double_quotes  task_resume_job       week4_T2
demo.txt               mmap_file.txt       task3_double_quotes.c task_resume_job.c     week4_T2.c
docs                   module_test          task3_double_quotes.c task_sigint_forward    week_5_double_quotes.c
dynamic_memory          obj                 task3_double_quotes.c task_sigint_forward.c.save
dynamic_memory.c        output.txt           task3_fork           task_sigint_forward.c
error.log
./assets:
architecture.png

./bin:
fork_exec

./docs:
README.md  architecture.txt  design.md  design_decisions.txt  execution_flow.txt  report.pdf  syscalls.txt

./include:
builtin.h  calculator.h  executor.h  parser.h  shell.h

./obj:
builtin.o  executor.o  main.o  parser.o  shell.o

```

```

./assets:
architecture.png

./bin:
fork_exec

./docs:
README.md  architecture.txt  design.md  design_decisions.txt  execution_flow.txt  report.pdf  syscalls.txt

./include:
builtin.h  calculator.h  executor.h  parser.h  shell.h

./obj:
builtin.o  executor.o  main.o  parser.o  shell.o

./scripts:
run.sh

./src:
builtin.c  executor.c  fork_exec.c  main.c  parser.c  shell.c

./tests:
test_executor.c  test_parser.c
alekhya@AlekhyaReddy:~/OSSP$ |

```